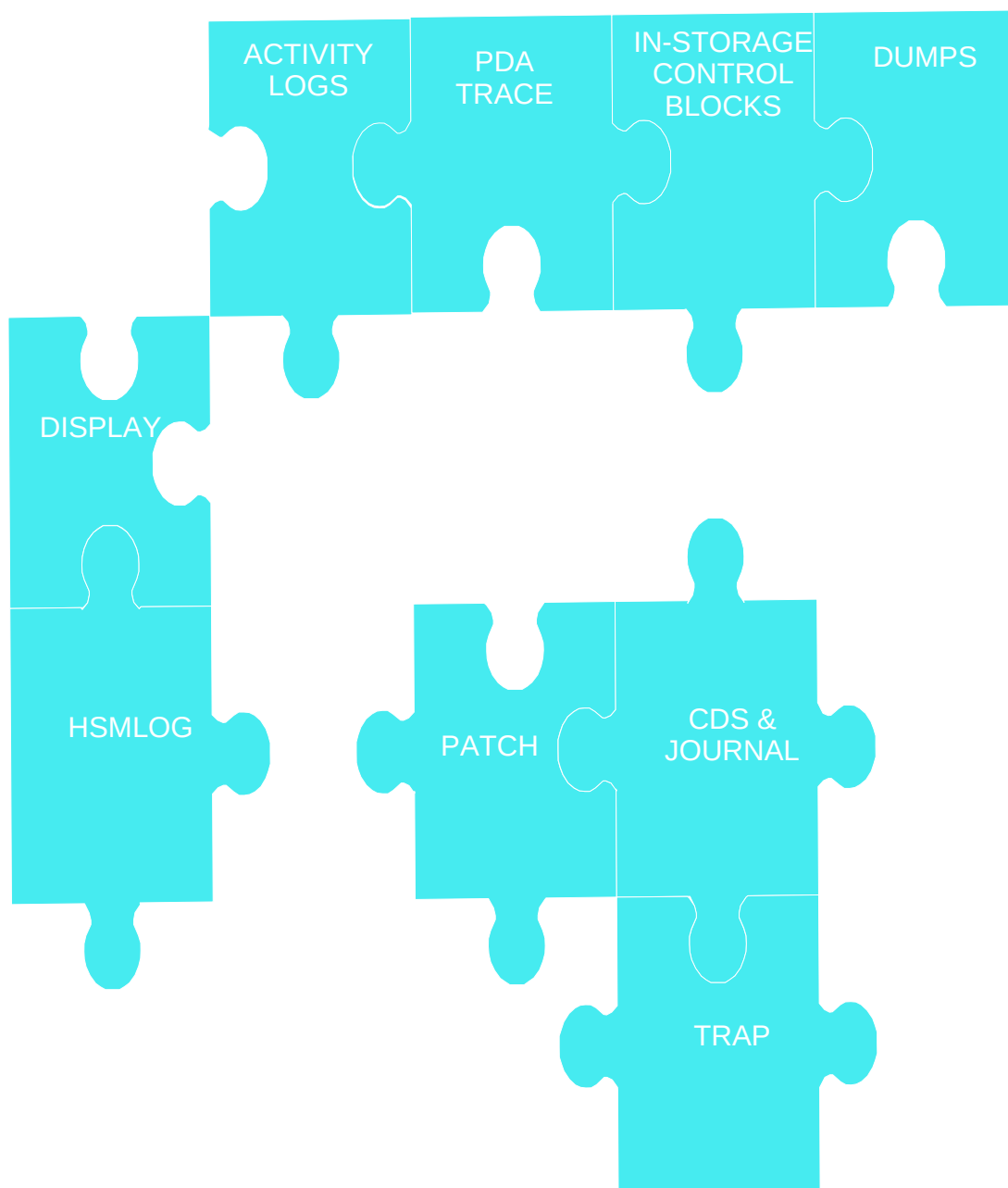


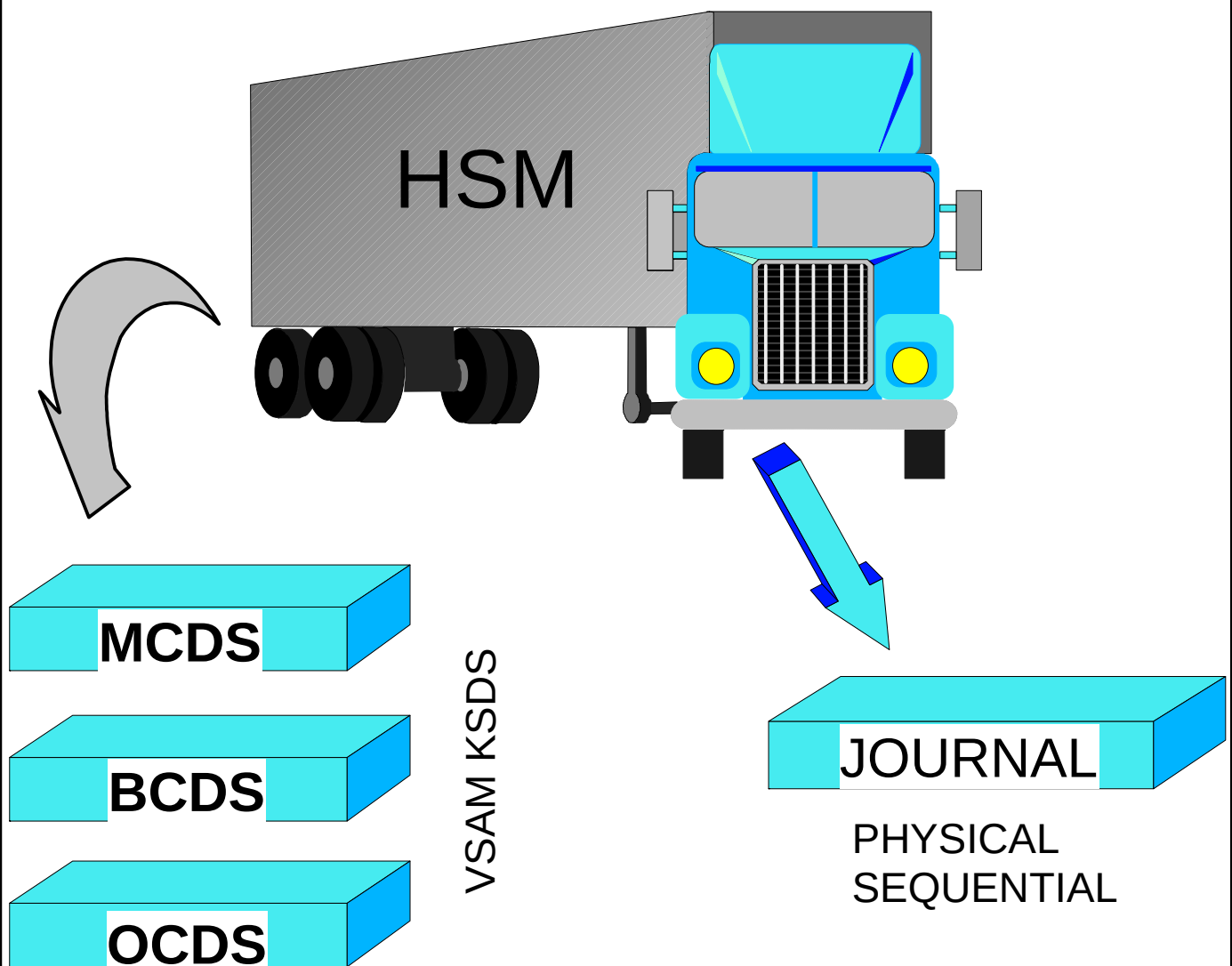




Objectives

- Understand the purpose of HSM control data set records.
- Understand what functions use which records, and when those records are created.
- Understand the inter-relationships between HSM control data set records.

HSM Control Data Sets



Migration Control Data Set - The MCDS contains a variety of different control data set records. Most records are migration function related, but the MCDS also contains authorized user records, volume records, statistic records and function control records.

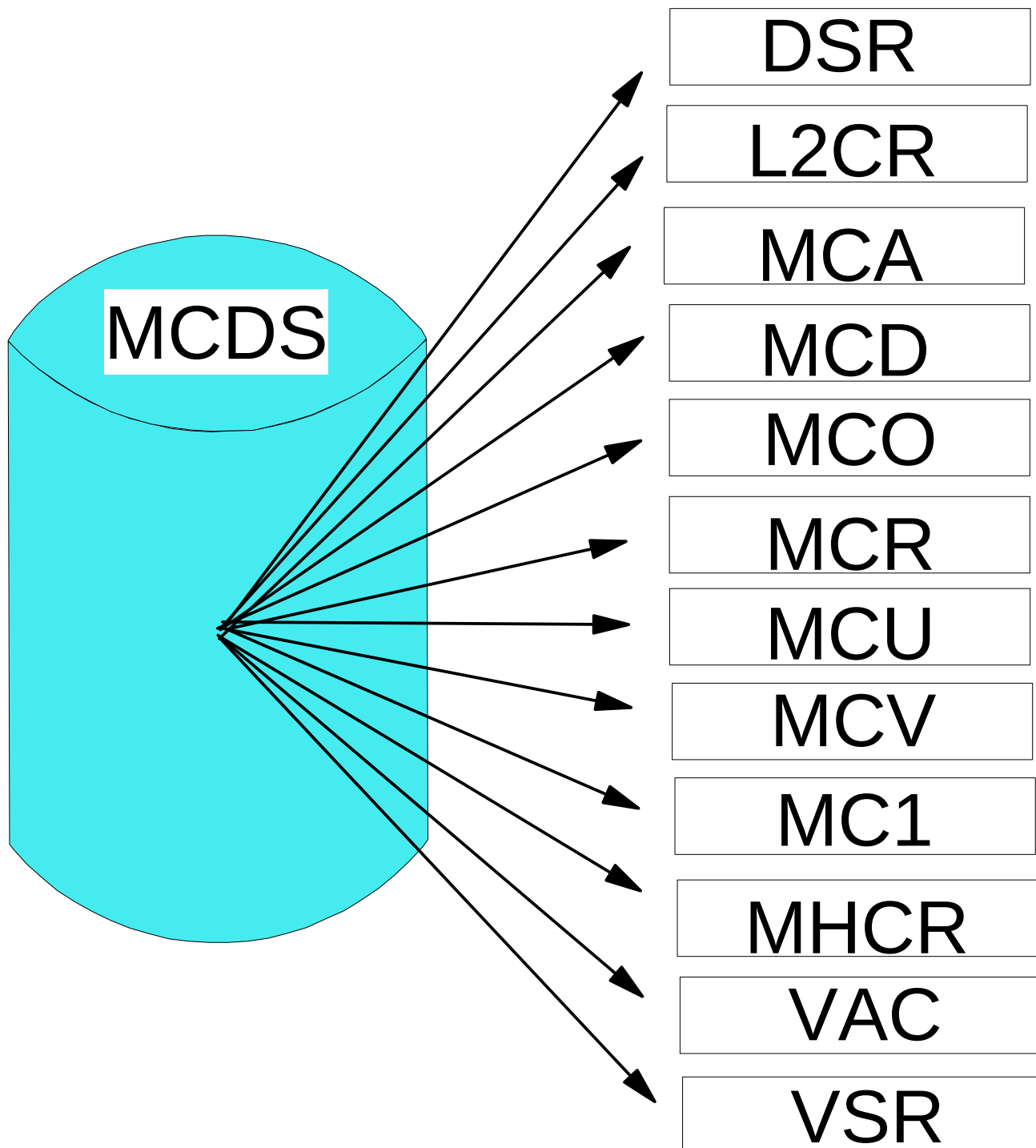
Backup Control Data Set - The BCDS contains backup and dump function related records. In addition, in DFSMSHsm, the BCDS contains the ABR record for keeping track Aggregate Backups used in ABARS.

Offline Control Data Set - The OCDS contains only a single record type, the Tape Table Of Contents (TTOC).

Journal - The HSM journal logs the creation, alteration, and deletion of various CDS records for all three control data sets. The Journal is a very important component as it is the method used to forward recover the CDS. The entries contain the CDS records themselves, preceded by a 12-byte header record (MLOG) that describes the type of transaction that was made to the CDS record.

Notes:

MCDS - Migration Control Data Set Records



DSR - Daily Statistics Record

L2CR - Migration Level 2 Control Record

MCA - Alias Entry Record

MCD - Data Set Record

MCO - VSAM Associations Record

MCR - Management Control Record

MCU - User Record

MCV - Volume Record

MC1 - Migration Level 1 Free Space Record

MHCR - Multiple-Processing-Unit Control Record

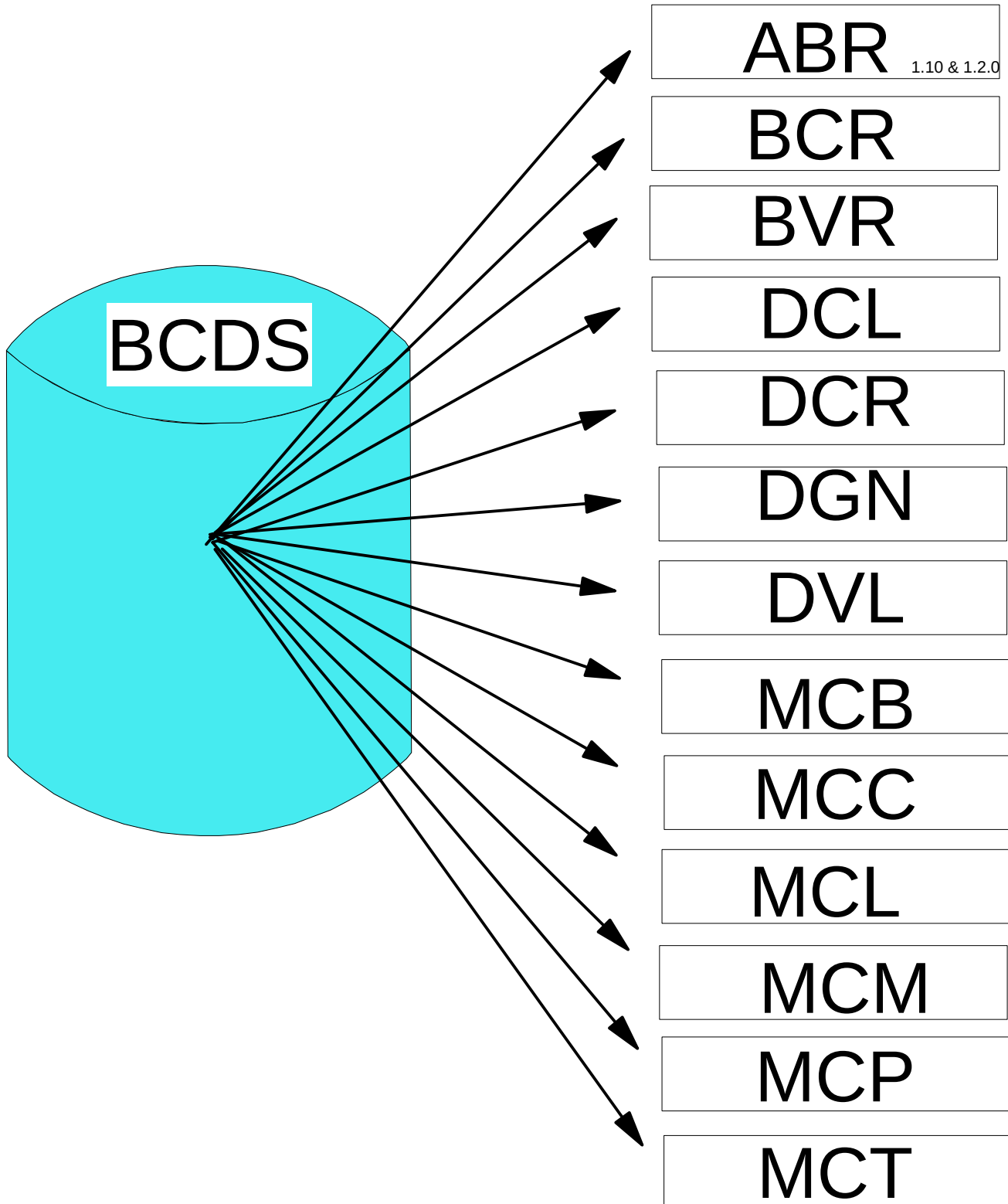
VAC - JES3 Volume Activity Count Record

VSR - Volume Statistics Record

Note: There is one MCR for each HSM image

Notes:

BCDS - Backup Control Data Set Records



ABR - Agregate Backup and Recovery Record

BCR - Backup Control Record

BVR - Backup Volume Record - subdivided into spill, unassigned, day of backup cycle (daily)

DCL - Dump Class Record

DCR - Dump Control Record

DGN - Dump Generation Record

DVL - Dump Volume Record

MCB - Data Set Record

MCC - Backup Version Record

MCL - Backup Changed Migrated Data Set Record

MCM - Backup Control Data Set Move Backup Version Record

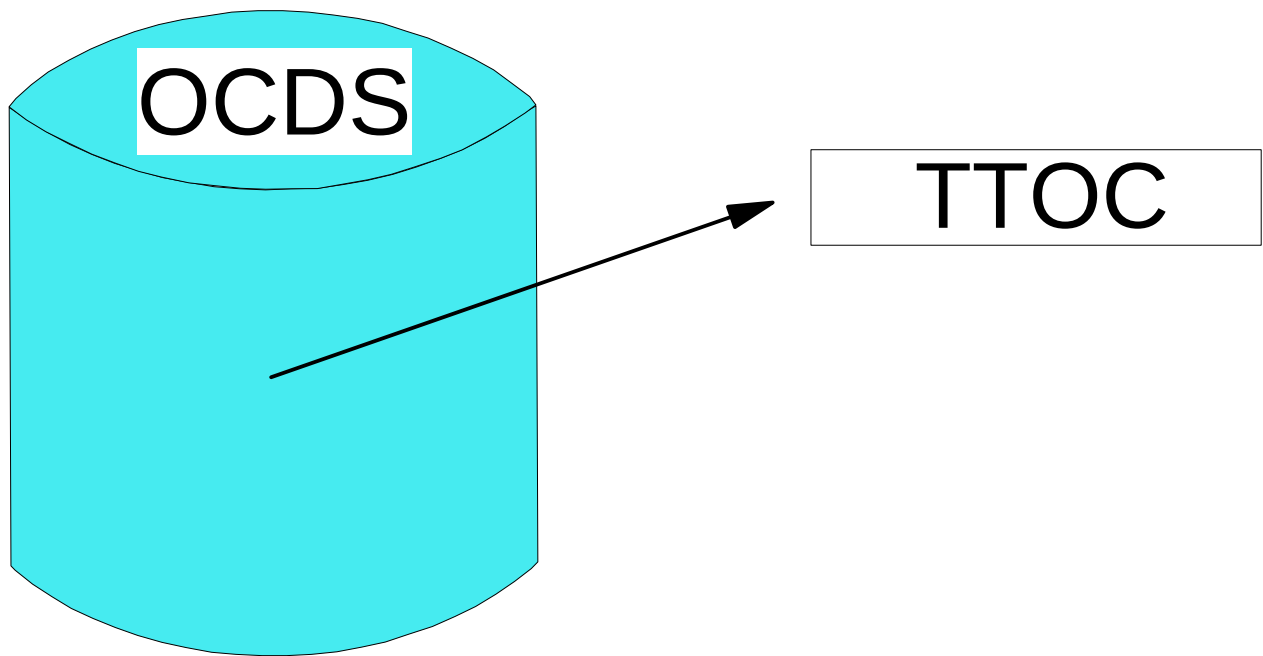
MCP - Eligible Volume Record

MCT - Backup Volume Record

Note: There is one BCR and DCR for each HSM image.

Notes:

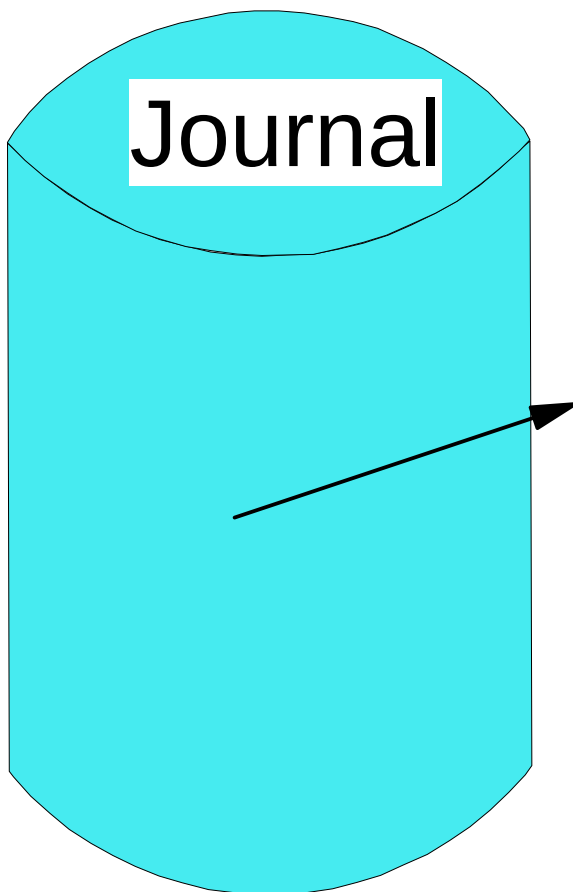
OCDS - Offline Control Data Set Records



TTOC - OCDS Tape Table Of Contents

Notes:

Journal Record



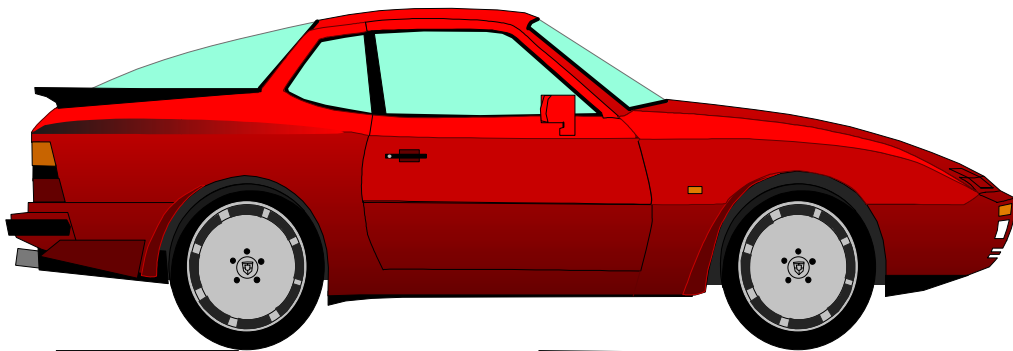
MLOG - Prefaces
CDS Records
in Journal

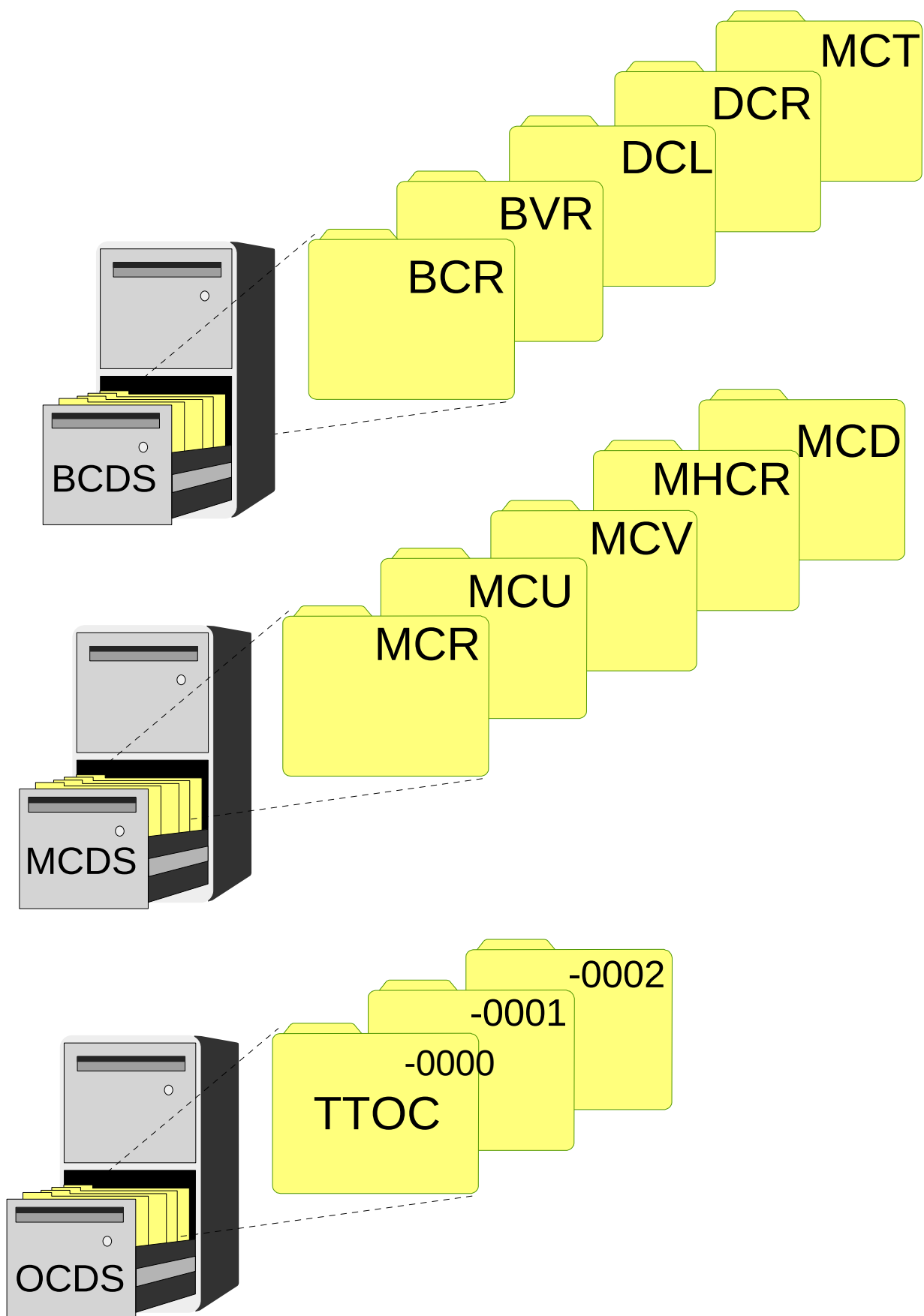
MLOG - Journal Record Header, precedes each CDS record written to the Journal. The MLOG is a 12-byte record containing day and time stamp information which can be cross referenced with the PDA trace JSYN entries. The MLOG also indicates if the Journal entry is for the creation of a new record or the update of an existing record. The date stamp (MLOGDATE) field is replaced during CDS backup processing with a sequence number used by the Enhanced CDS Recovery procedure.

Notes:

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Command-Driven CDS Records





BCDS

BCR - Contains information pertaining to the backup functions. In addition, there is some additional information about the EXPIREBV function.

BVR - Contains volume information all SPILL, DAILY, or UNASSIGNED backup volumes.

DCL - Created as a result of the DEFINE DUMPCLASS command.

DCR - Contains information pertaining to the DUMP function.

MCT - Created for all backup volumes that have an ADDVOL statement in the ARCCMDxx PARMLIB member.

MCDS

MCD - SETMIG command

MCR - Contains information pertaining to the migration functions. Cycles, start times, dates, times, etc....

MCU - Describes an authorized DFHSM user. This record is created in response to the AUTH command in your ARCCMDxx PARMLIB member..

MCV - Created for all primary or migration volumes when there is an ADDVOL statement in the ARCCMDxx PARMLIB member. Also, internally created for all SMS-managed volumes.

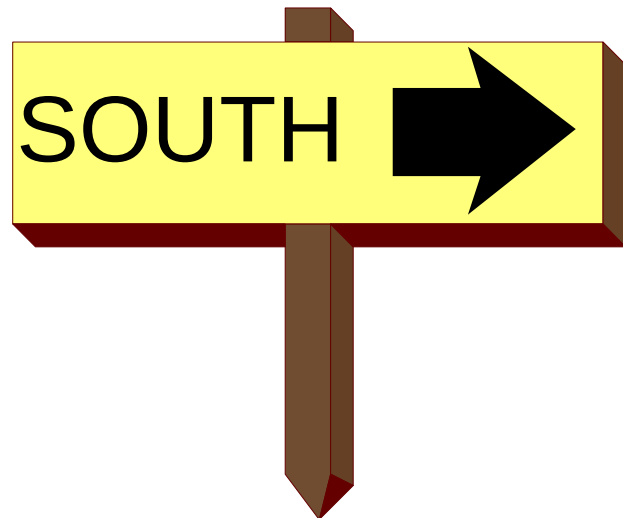
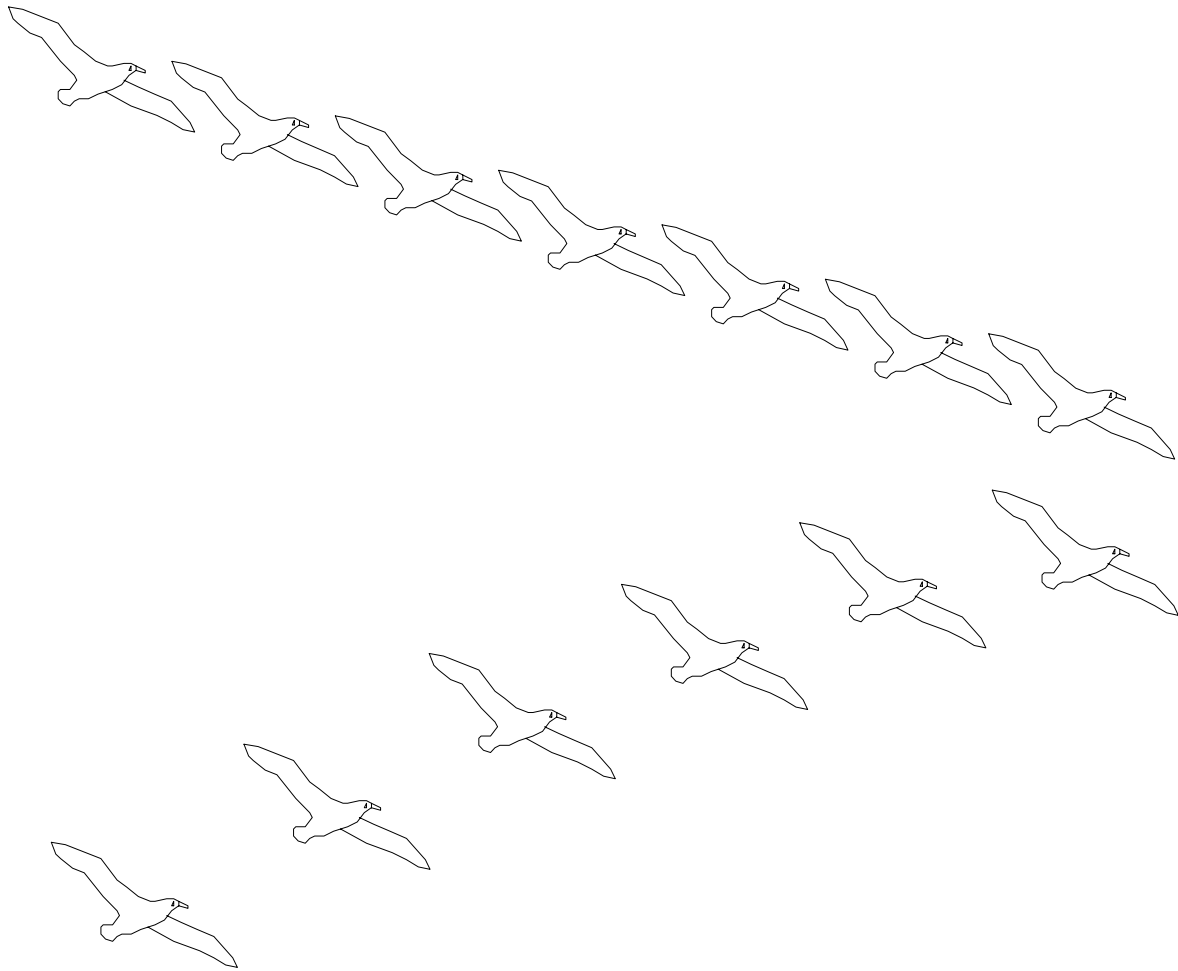
MHCR - Contains information pertaining to the Control Data Sets.
NOTE: The MHCR is not recorded in the journal.

OCDS

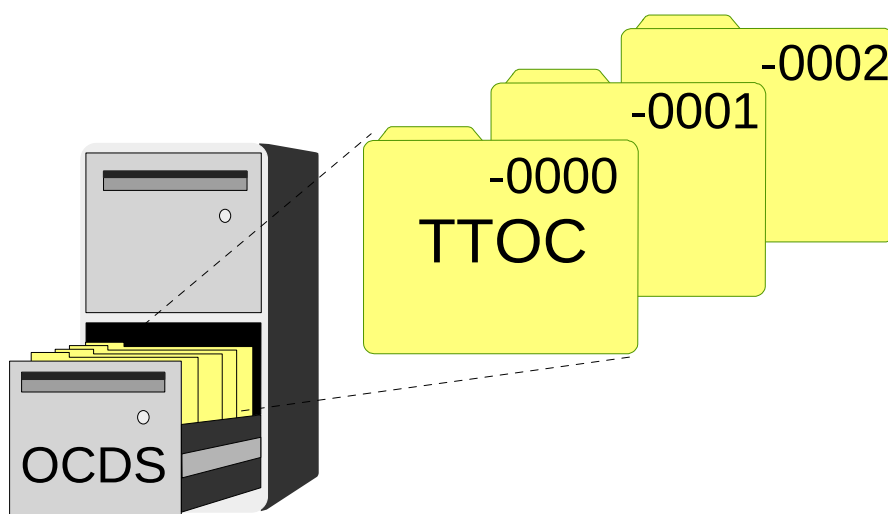
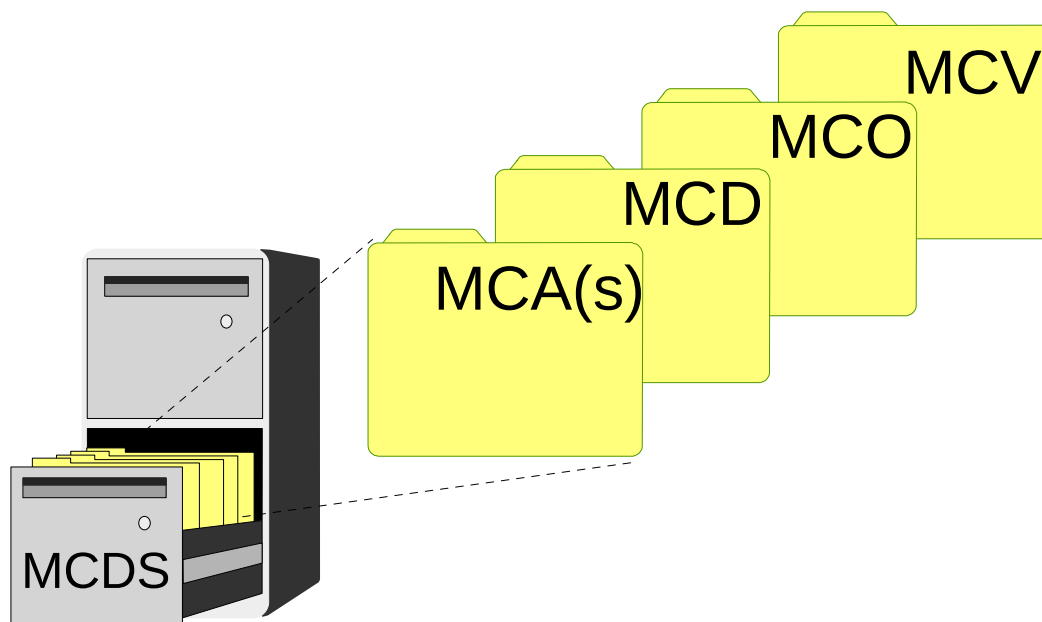
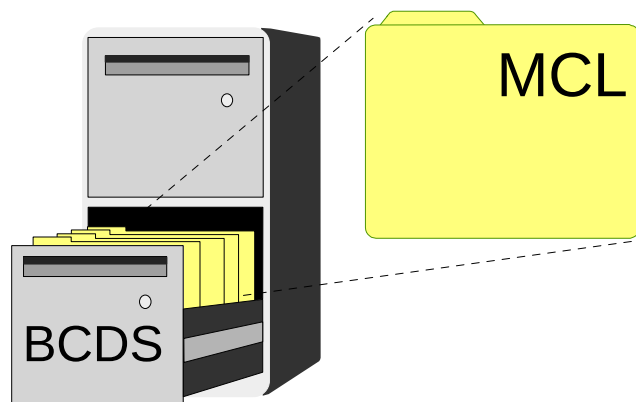
TTOC - Contains information pertaining to tape backup and ML2 tape migration data sets.

Notes:

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Records
Created During Migration



BCDS

MCL - Created for each data set in need of backup at the time of migration to ML1.

MCDS

MCA - Created for all data sets migrating from L0 volumes except SDSP targeted data sets. Also created for data sets migrating from ML1 to ML2 if coming FROM an SDSP. For VSAM data sets supported by automatic migration, an MCA record is created for each component.

MCD - Created for all migrated data sets.

MCO - Created for all migrated VSAM data sets supported by automatic migration.

MCV - Created for new tapes added to HSM's inventory during migration to tape.

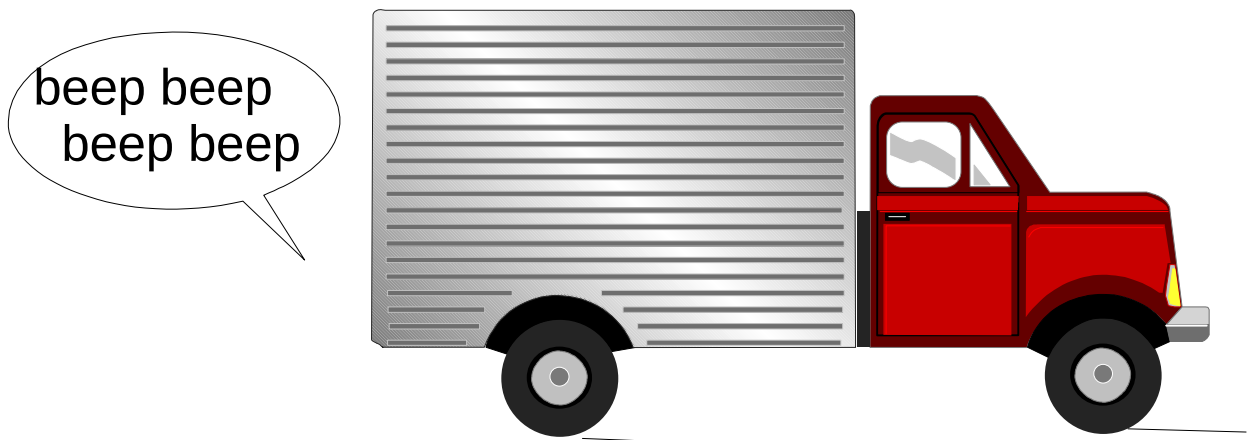
OCDS

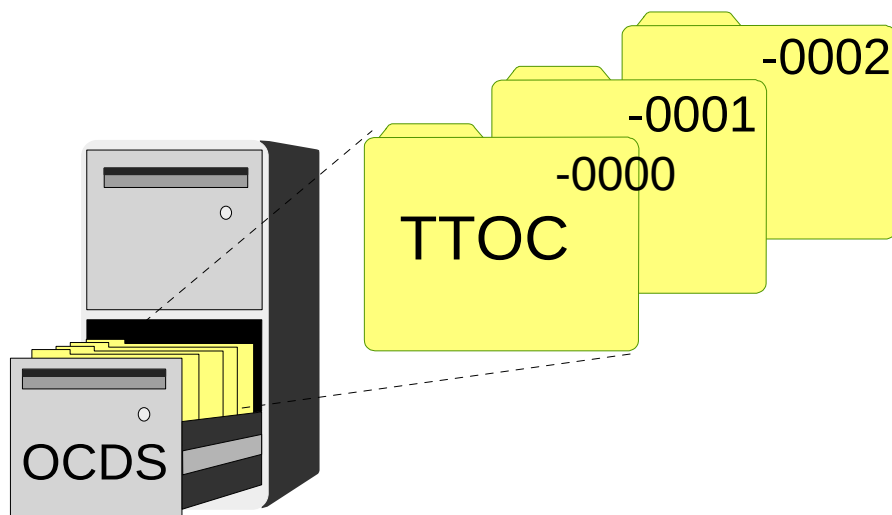
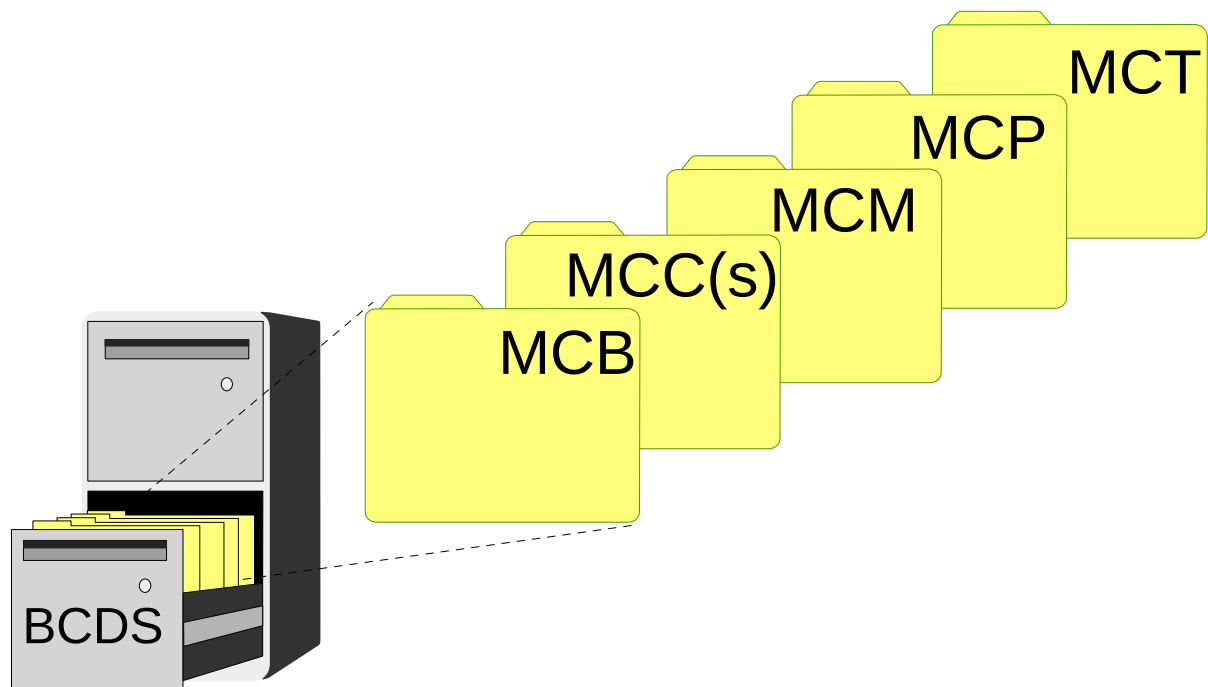
TTOC - Created for new tapes added to HSM's inventory during migration to tape. Each extension contains up to 33 data sets. The number of extensions is determined by the number of data sets on the tape. The TTOC extensions are created when the tape begins to fill. An extension is created when the Base TTOC reaches a limit of 33 data sets.

Notes:

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Records Created During Backup





BCDS

MCB - Created when a data set is backed up for the first time. If an MCB record exists, HSM will update it with current backup version information.

MCC - Created for each backup version up to a maximum of 13.

MCM - Created for data sets that are backed up using the BACKDS or HBACKDS command.

MCP - Created for each primary volume and updated as needed. Tracks information pertaining to a primary volume's backup history.

MCT - Created for new tapes added to HSM's inventory during backup to tape.

OCDS

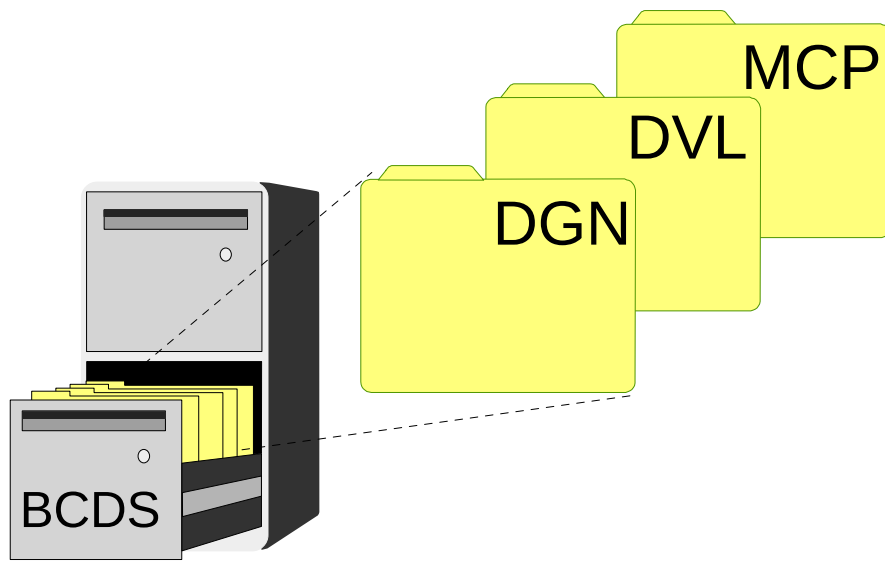
TTOC - Created for new tapes added to HSM's inventory during backup to tape.

Notes:

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Records Created During Dump





BCDS

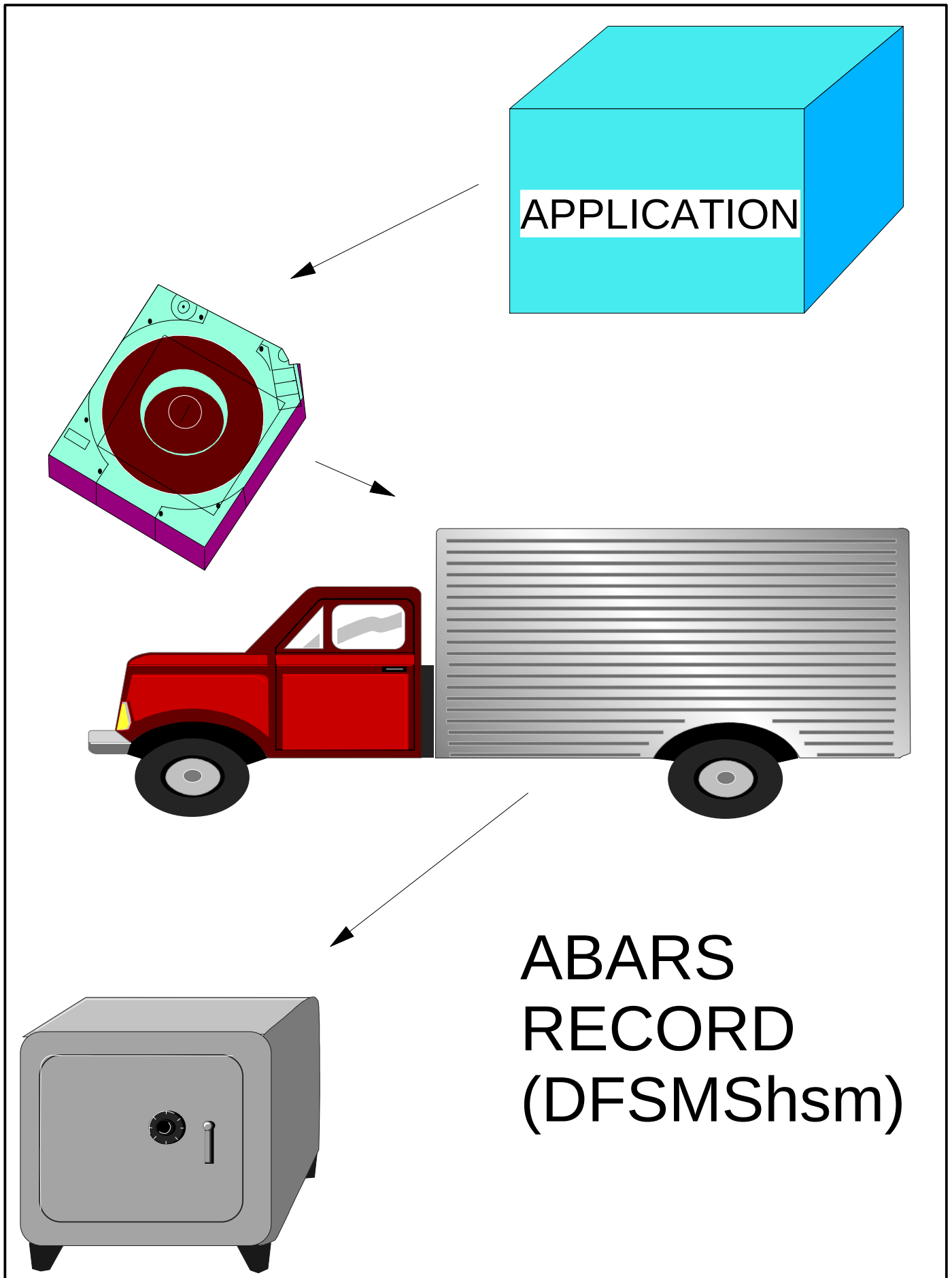
DGN - Describes a dump generation and all of its associated dump volumes.

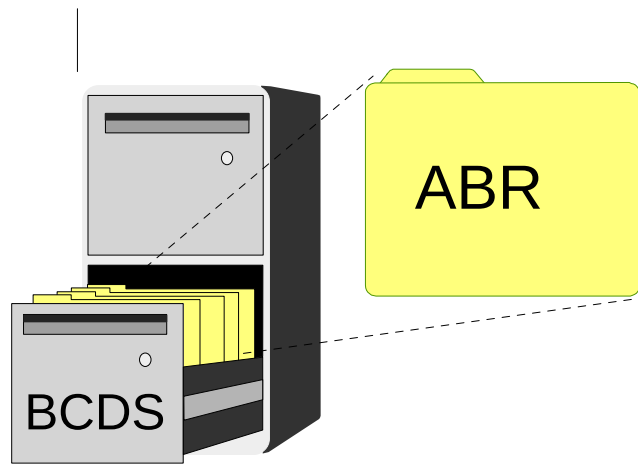
DVL - Contains information about an individual volume within a dump generation.

MCP - Created for each volume that is dumped and updated as needed. Contains information pertaining to a primary volume's dump history.

Notes:

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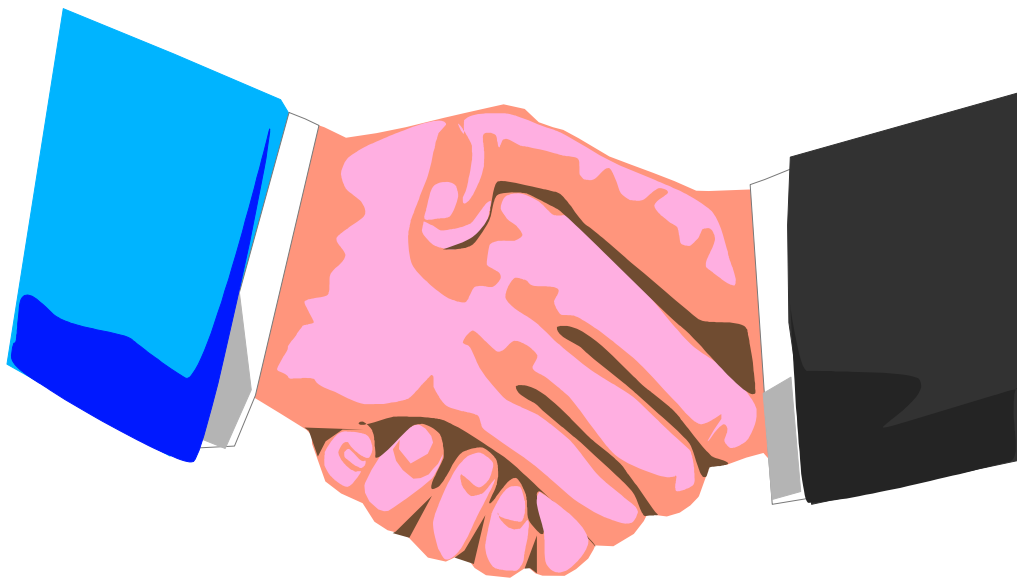
BCDS

ABR - This record contains information about a specific aggregate backup version. The ABR record also contains information about the last ARECOVER processed using this version. EXPIREBV references this record in order to determine when old ABR records should be expired. This record only exists with DFSMSHsm.

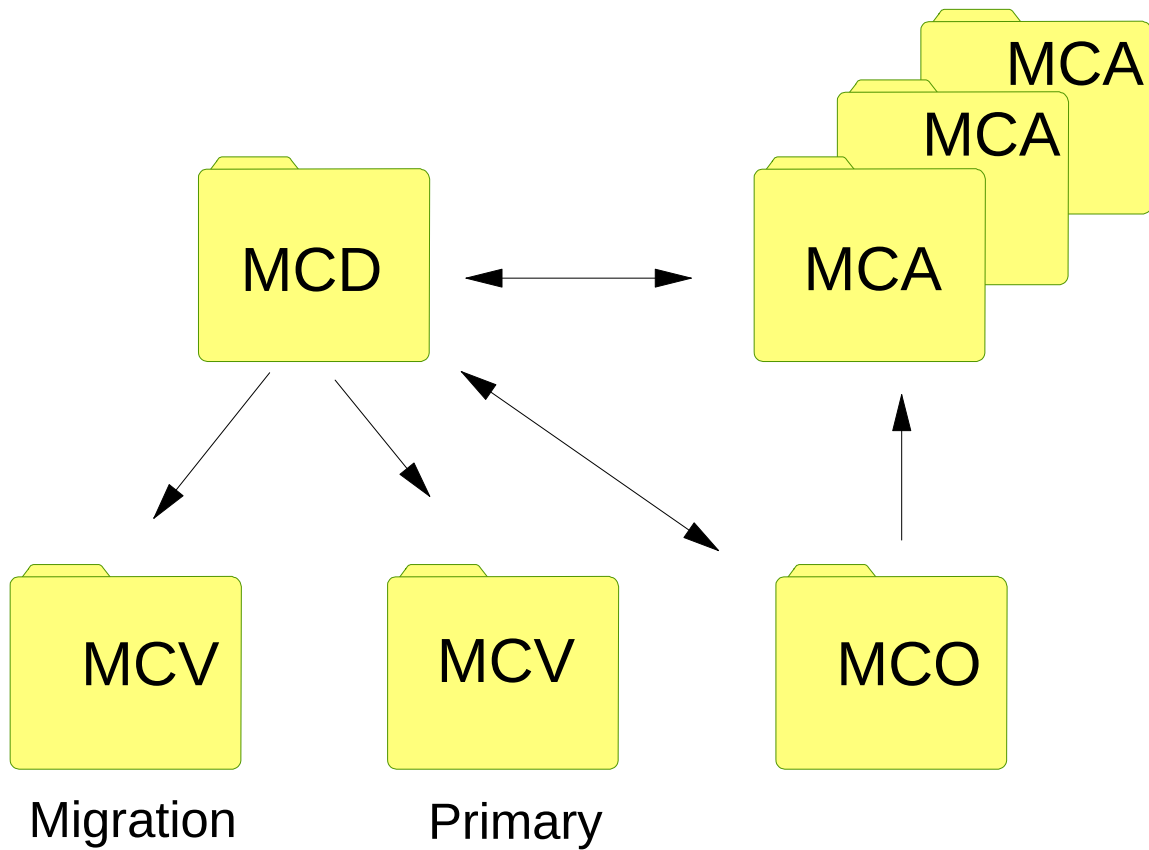
Notes:

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CONTROL DATA SET RELATIONSHIPS



Migration Record Relationships



MCD - Key is the original data set name.

- MCDMCANM contains the key to the MCA record.
- MCDVSN contains the key to the MCV record of the migration volume.
- MCDFRVSN contains the key to the MCV record of the primary volume.

MCA - Key is the migrated data set name or VSAM component name.

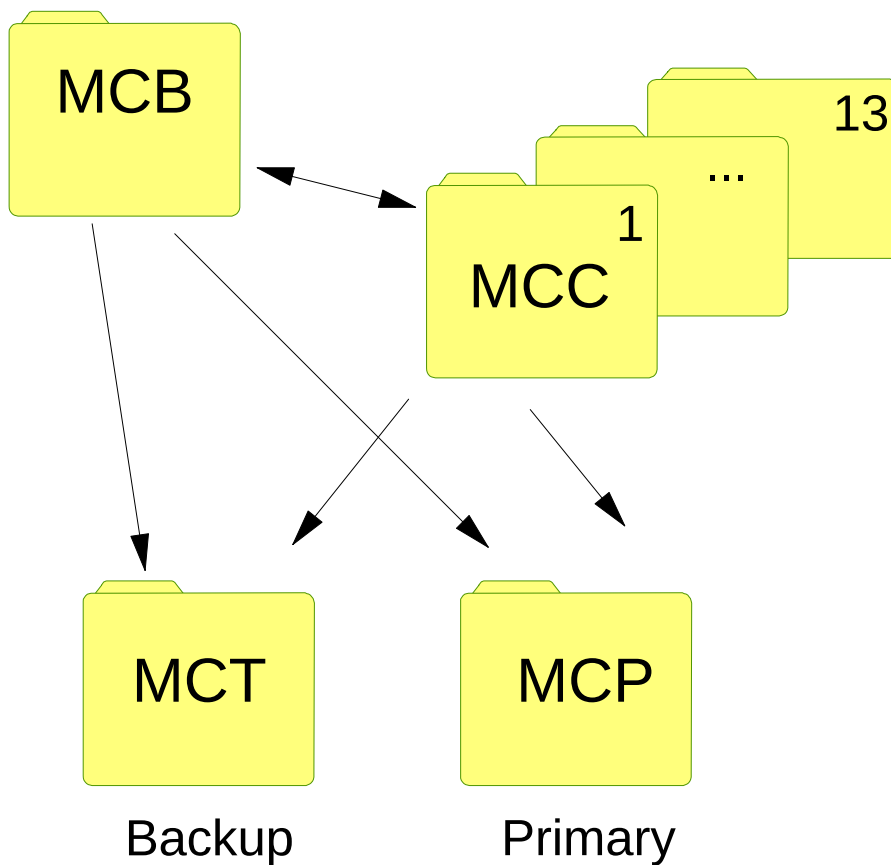
- The MCA record is used by the RECYCLE, AUDIT, and FREEVOL commands.
- Used during ML1 to ML2 migration.
- MCAINTNM contains the key to the MCD record.
- Allows implicit recall by component name of data set whose key is the cluster name.

MCO - Key is the migrated data set name.

- Contains information specific to VSAM data sets.
- MCOORGNM contains the key to the MCD record.

Notes:

Backup Record Relationships



MCB - Key is original data set name.

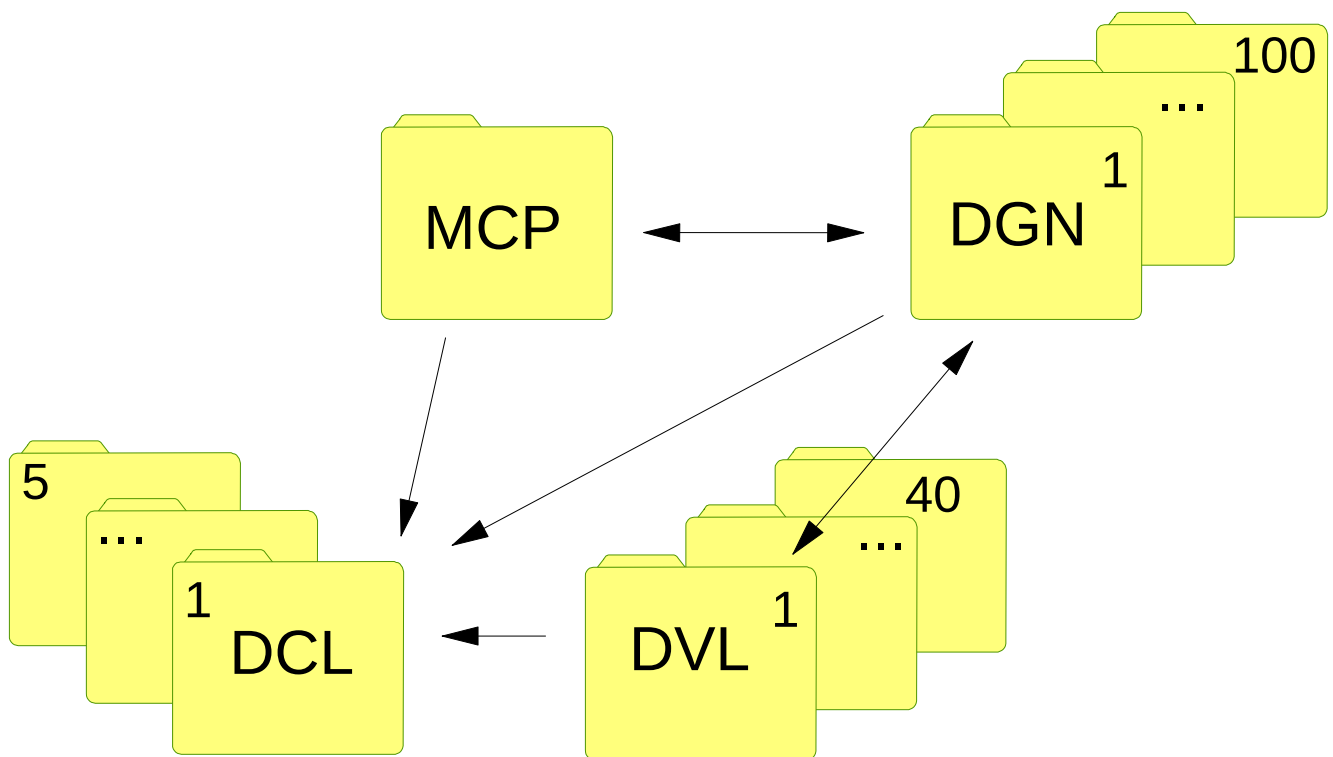
- MCBBUI begins an array that points to all associated MCC records.

MCC - Key is backup version name, one for each of up to 13 backup versions.

- MCCADSN contains the key of the MCB record
- MCCVSN contains the key of the MCT record.
- MCCFRVOL contains the key of the MCP record for the primary volume.

Notes:

Dump Record Relationships



MCP - Key is the primary volume volser.

- MCPGNNM begins an array of 100 DGN pointers/keys
- MCPLDCDT begins an array of 5 DCL pointers/keys.

DGN - Key is the L0 volser + time stamp + date stamp

- DGNML1VS contains key to MCV of L0 volume.
- DGN_INFO begins an array describing each of the dump copies that make up this generation, including keys for each DVL record.

DVL - Key is dump volume volser.

- DVLCLAS contains the key of the DCL record.
- DVLDGNKY contains key of the DGN record for the generation that this dump volume belongs to.

Notes:

Diagnostic Value of the CDS and Journal

The CDSs are the virtual backbone of HSM. All HSM activity is either tied to or recorded in the control data sets. Everything from when Primary Space Management last ran on a volume to when a data set was last backed up is recorded in the CDSs. This inter-relationship between HSM and the CDSs is what makes them such a valuable source of diagnostic information.